Program 3:
Bit stuffing and Character(Byte Stuffing)

AIM: Write a program to implement stuffing techniques. i. Bit-stuffing ii. Character stuffing

Description:

Bit Stuffing is a framing technique used in the Data Link Layer to ensure that the special flag pattern used to mark the beginning and end of a frame does not appear in the transmitted data. In this technique, whenever the sender encounters five consecutive 1s in the data stream, it automatically inserts a 0 bit. The receiver removes the inserted 0 during destuffing to recover the original data. This method is commonly used in bit-oriented protocols such as HDLC.

Character Stuffing (also known as Byte Stuffing) is a framing technique in which special control characters are used to indicate the start and end of a frame. If the same control or flag character appears in the data, the sender inserts an Escape (ESC) character before it to distinguish it from the frame delimiter. At the receiver, the ESC character is removed during destuffing to reconstruct the original data. This technique is commonly used in character-oriented communication protocols.

Program Bit Stuffing:

```
#include <stdio.h>
#include <string.h>
int main() {
    char data[100], stuffed[200];
    int i, j = 0, count = 0;
    printf("Enter the bit stream: ");
    scanf("%s", data);
    for(i = 0; data[i] != '\0'; i++) {
        stuffed[j++] = data[i];

        if(data[i] == '1') {
            count++;
            if(count == 5) {
                stuffed[j++] = '0';   // Stuff a 0
                count = 0;
            }
        } else {
            count = 0;
        }
    }
    stuffed[j] = '\0';
    printf("\nOriginal Data : %s", data);
    printf("\nStuffed Data  : %s\n", stuffed);

    return 0;
```

```
}
```

OUTPUT:

```
Enter the bit stream:
1111101111110

Original Data : 1111101111110
Stuffed Data  : 111110011111010
```

Byte Stuffing Program:

1. **F** is used as the **Frame Flag** (start/end of the frame).
2. **E** is used as the **Escape (ESC)** character.
3. If **F** or **E** appears in the data, an **E** is inserted before it.

```
#include <stdio.h>
int main() {
   char data[100], stuffed[200];
   int i, j = 0;

   printf("Enter the data: ");
   scanf("%s", data);

   stuffed[j++] = 'F';   // Start Flag

   for(i = 0; data[i] != '\0'; i++) {
      if(data[i] == 'F' || data[i] == 'E') {
         stuffed[j++] = 'E';   // Escape Character
      }
      stuffed[j++] = data[i];
   }

   stuffed[j++] = 'F';   // End Flag
   stuffed[j] = '\0';

   printf("\nOriginal Data : %s", data);
   printf("\nStuffed Data  : %s\n", stuffed);

   return 0;
}
```

OUTPUT:

```
Enter the data: ABFECD
Original Data : ABFECD
Stuffed Data  : FABEFEECDF
```

1 Marks Questions

1. What are the design issues of the Data Link Layer?

Ans: The main design issues of the Data Link Layer are:Providing services to the Network Layer ,Framing ,Error Control ,Flow Control and Medium Access Control (MAC).

2. What is framing? Why is it required?

Ans: Framing is the process of dividing a stream of bits into identifiable units called **frames**.It is needed to identify the beginning and end of data , to enable error detection and retransmission and to synchronize sender and receiver.

3. List the different framing techniques.

Ans: 1. Byte count. 2. Flag bytes with byte stuffing. 3. Flag bits with bit stuffing. 4. Physical layer coding violations.

4. What is Character Stuffing (Byte Stuffing)?

Ans: If a flag byte (FLAG) appears in the data, it may be mistaken for a frame boundary. To avoid this, the sender inserts an escape byte (ESC) before every flag byte in the data. The receiver removes the escape bytes before delivering the data to the network layer. This technique is called Byte Stuffing.

5. What is Bit Stuffing?

Whenever the sender detects **five consecutive 1s** in the data, it **inserts a 0 bit** automatically. This prevents the flag pattern from appearing within the data and ensures correct frame identification. This technique is called Bit Stuffing.

6. What is the flag pattern used in Bit Stuffing?

The flag pattern is: **01111110 (0x7E)** It marks the **beginning and end of a frame**.

7. What happens if the character count field is corrupted?

If the character count field is corrupted, the receiver calculates an incorrect frame length, resulting in **loss of synchronization** and incorrect frame interpretation.